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Pesticide exposure, tobacco use, poor self-perceived health and presence of chronic disease are determinants of depressive symptoms among coffee growers from Southeast Brazil

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ABSTRACT

The lifestyle and other factors associated with the appearance of several health conditions that affect quality of life in rural zone is an issue that has been increasingly explored. Brazil is the largest coffee-producing nation in the world and has been a considerable consumer of pesticides since 2008. The aim of the present study was to investigate factors that could be contributing to the appearance of depressive symptoms in rural workers. Two hundred twenty male volunteers from nine cities in Southeast Brazil completed the Beck Depression Inventory-II (BDI-II) questionnaire about depressive symptoms and provided other information about socio-demographic characteristics and additional confounding factors. The adjusted multivariate logistic analysis demonstrated that pesticide exposure, tobacco use, poor self-perceived health and the presence of chronic disease contribute as risk factors for the appearance of depressive symptoms at a level above ups and downs considered normal in the BDI-II. This survey contributes to the search for solutions to improve quality of life and mental health in the rural living to the extent that social determinants of depression are being investigated.

1. Introduction

The lifestyle and factors associated with the appearance of several conditions that affect the quality of life in the country is an issue that has been increasingly explored (Gambin et al., 2015; Pignatti and Castro, 2008, 2010; Tavares et al., 2015). Rural myths could make us believe that country life has only health benefits; however, studies have shown that rural living is far from the tranquility of “rural existence” (AIHW, 1998; Hansen, 1987).

Work and life conditions are very precarious, and the monoculture practiced in the world of agribusiness is an important source of distress and illness (Scopinho, 2010). Brazil is the largest coffee-producing nation in the world, and historically, for over 150 years, it has been the highest global producer of coffee beans (USDA, 2016). The Southeast region included in the present study belongs to the state of Espírito Santo, where coffee growing is the main and most traditional agricultural activity. Espírito Santo represents the second largest coffee producer in Brazil and the first worldwide in *Robusta* production (Cetcaf, 2014; USDA, 2016).

As a consequence of peculiar life style, researches have described the presence of depressive symptoms in agricultural workers compromising their mental health (Phillips and Deshpande, 2016; Rayens and Reed, 2014). In the present study, we investigated determinant factors that could be contributing to the appearance of these symptoms, that substantially compromise quality of life (Cruz et al., 2010), and, depending on the severity and intensity, its impact on general welfare can be up to 23 times greater than other physical diseases (Williams et al., 1995). Depression changes the way one sees the world, perceives reality, comprehends things and expresses emotions. Therefore, it is considered a disease that affects the entire body and human experience without fragmentation between the biological, mental and social issues. Risk factors were evaluated through a multivariate logistic model and these data may contribute to the search for public health policies to improve mental and general health of rural residents.

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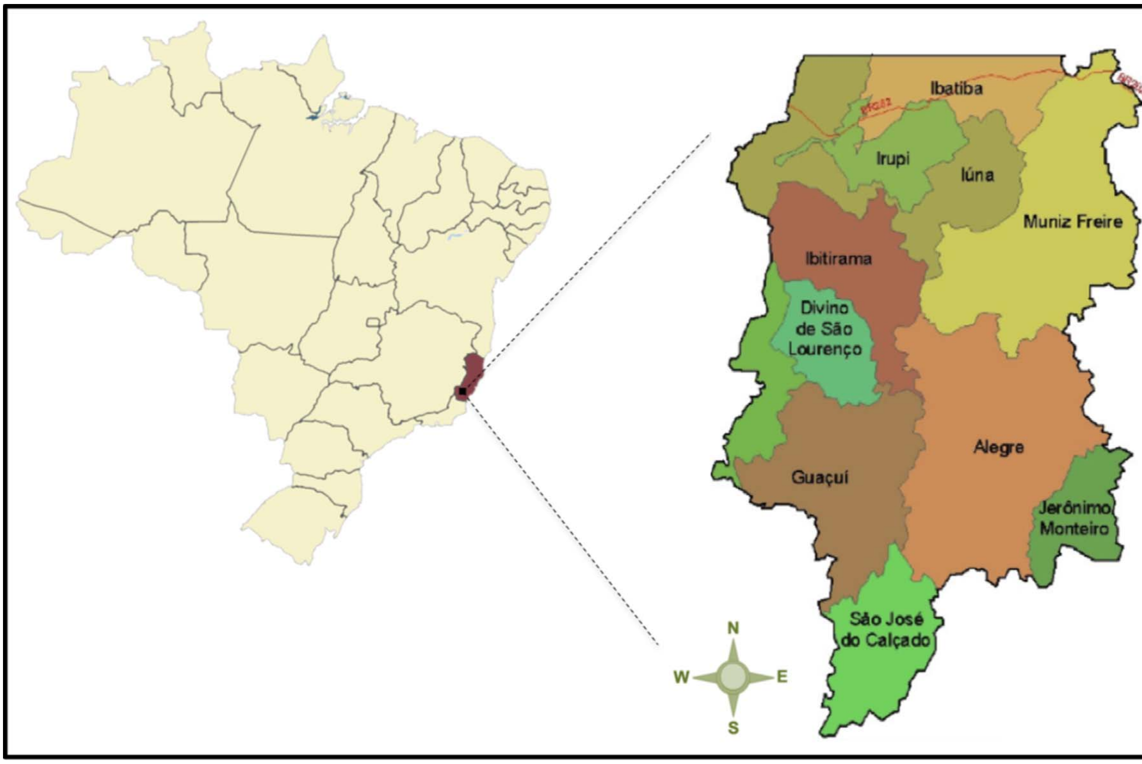

Fig. 1. Map of the Caparaó Capixaba consortium included in this study – the greatest *Robusta* producer worldwide.

2. Materials and methods

2.1. Subjects and ethics statement

This study included two hundred twenty (N = 220) male volunteers from 18 to 65 years of age. All participants are rural workers from nine cities (Alegre, Dores do Rio Preto, Guaçuí, Ibatiba, Ibitirama, Irupi, Iúna, Muniz Freire and Jerônimo Monteiro,) of Southeast Brazil that together form the Caparaó Capixaba Consortium (Fig. 1), the greatest *Robusta* producer worldwide. After a non-successful pilot testing within a random sample, the sampling was performed by convenience mainly because of the difficulty of contact by phone to schedule the meeting with them and the lack of participant availability once they live in remote areas with well-established routines. The number of participants in each city was calculated according to the percentage of the number of rural residents from that city and, in the total sample, the number was based on the rural population of the whole region, a total of 63,389 residents, according to data from the Brazilian Institute of Geography and Statistics (IBGE, 2010), considering a minimum prevalence of depressive symptoms of 14% according previous literature (Gomes-Oliveira et al., 2012; Silva et al., 2014), with a sampling error of 5%, a 95% confidence interval and 1.2 of design effect. It was selected only one participant per family to minimize homogeneity. This survey was conducted according to the ethical principles established by the Ethics Committee for Research at the Center of Health Sciences, Federal University of Espírito Santo, Brazil. All participants signed the informed consent forms. This research is part of a project approved by this ethics committee under registration #1.634.021.

2.2. Data collection

The secretary of the health department from each city was contacted to provide the consent letter. Then, the head of each community was contacted to schedule a meeting in which our group would explain the project to associated subjects and obtain their consent to participate. A few days later, the group returned to collect the socio-demographic data, information about pesticide exposure (yes or no, according to any type of direct contact with pesticides) and other habit information like alcohol and tobacco use. Besides this structured questionnaire, it was additionally administered the validated Beck Depression Inventory-II questionnaire which gave rise to the primary outcome variable. The forms were administered by trained students and professionals of the University, while the Beck form was administered by only one or two trained people to minimize bias.

2.3. Beck Depression Inventory-Second Edition (BDI-II)

The BDI-II is not a depressive disorder diagnosis tool; instead, it is a self-report instrument that assesses the presence and severity of depressive symptoms in normal and psychiatric populations. It consists of 21 items that are rated on a 4-point scale, ranging from 0 to 3, with higher scores indicating more severe symptoms of depression (Jackson-Koku, 2016). It was already shown that the BDI-II is reliable and valid for measuring depressive symptomatology among Portuguese-speaking Brazilian non-clinical populations. The intraclass correlation coefficient of the BDI-II was 0.89, and Cronbach's alpha coefficient of internal consistency was 0.93 (Gomes-Oliveira et al., 2012).

2.4. Factors included in the analysis (variables)

Depressive symptoms were treated as the dependent variable. For statistical analysis, it was categorically coded as follows: low Beck score — up to 10 for the final score — according to the consensus parameter for ups and downs is considered normal for both rural and Brazilian populations (Iatchac et al., 2010; Levandovski et al., 2011), and high Beck score (total score greater than 10). The independent variables were as follows: age, marital status, Brazilian racial categories (self-reported), years of schooling, economic status of family (one salary refers to U$ 282), alcohol use, tobacco use, combined alcohol and tobacco use, pesticide exposure, type of pesticide, self-perceived health and presence of chronic disease.

All variables are reported in Fig. 2.

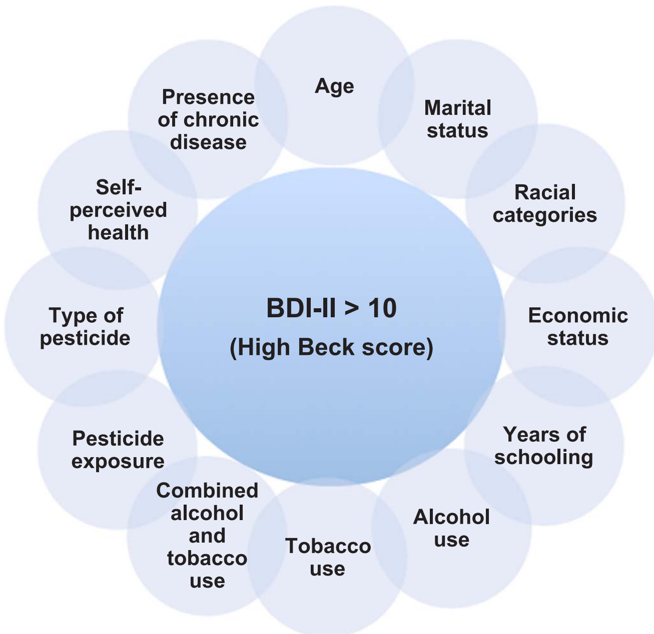

Fig. 2. Dependent and independent variables considered in the multivariate logistic analysis.

2.5. Statistical analyses

The univariate analysis was performed to evaluate the association between high Beck score and their possible risk factors, followed by multivariate logistic regression. In the regression analysis, the multivariate model was adjusted with all significant variables considering $p < 0.02$, surviving at the final model only those variables with significance defined as $p < 0.05$. All analyses were performed using the Stata® statistical software package version 14.

3. Results

The associations between depressive symptoms and each independent factor after univariate analysis, are shown in Table 1. Briefly, the analysis suggested that pesticide exposure, fungicide and insecticide use, poor self-perceived health and the presence of chronic disease were significantly (*p < 0.05, **p < 0.005) associated with a high Beck score.

3.1. Pesticide exposure

The main type of pesticide used by the participants in this study was the herbicide glyphosate (77.3%). However, others were also substantially related, including flutriafol, cyproconazole and thiamethoxam. Their characterization is shown in Table 2. The univariate analysis showed that exposure to these pesticides, except glyphosate alone, is suggested to be a risk factor for the development of depressive symptoms. However, as the glyphosate was extensively related to the symptoms, this predictive analysis of the relationship between the type of pesticide to which they were exposed and high Beck scores was not exclusively performed; instead, it was always associated with the use of glyphosate. Of all participants exposed to flutriafol ($N = 62$), 92% also used glyphosate; 87% of those exposed to cyproconazole also used glyphosate, while 88% of those exposed to thiamethoxam also used glyphosate. This result may be interpreted as follows: the use of glyphosate alone was not a significant risk factor, but the use of flutriafol, cyproconazole or thiamethoxam, when associated with glyphosate exposure, was a significant risk factor for the appearance of depressive symptoms. Analysis of exposure to the low amounts of these three types of pesticides alone, with no combined exposure to glyphosate, was performed, and no significant result was found (not shown).

Multivariate logistic regression adjusted by pesticide exposure, type of pesticides, health conditions and tobacco use was performed and predicted risk factors for depressive symptoms are shown in Table 3 (pseudo $R^2 = 0.1024$; prob > chi^2 = 0.0001).

4. Discussion

We demonstrated, through univariate analysis, that exposure to pesticides, the application of fungicides and insecticides associated with glyphosate in the tillage, poor self-perceived health and the presence of chronic disease are associated with the appearance of depressive symptoms at a level above ups and downs considered normal in the BDI-II. Adjusted multivariate analysis showed that pesticide exposure, tobacco use, poor self-perceived health and presence of chronic disease were determinant to a high Beck score.

Mood disorders, including depression, are the most prevalent psychiatric disorders in current society. Major depression affects approximately 16% of people once during their lifetime (Kessler et al., 2005). All symptoms of these disorders are collectively called ‘depressive syndrome’ and they are characterized by anxiety, feelings of guilt, long-lasting depressed mood and recurrent thoughts of death and suicide (Nestler et al., 2002). Pesticide exposure is widely known to be a relevant environmental health issue in rural communities with agricultural activity (Coronado et al., 2011; Kim et al., 2017; Li, 2018; Souza et al., 2017; Valcke et al., 2017). Several studies have demonstrated the association between the toxicity of pesticides and the onset of psychiatric conditions, especially depression, among rural workers and residents (Beseler and Stallones, 2013; London et al., 2012a; Stallones and Beseler, 2016; Wesseling et al., 2010). Studies from Brazil (Meyer et al., 2010), Spain (Parron et al., 2011) and the USA (Beseler and Stallones, 2008) identified the relationship between pesticide exposure and psychiatric disorders. The indiscriminate use of pesticides is leading to concerns about their potential acute and chronic effects in general health and especially in mental health. In Brazil, a country with many rural workers, this problem is of great relevance because Brazil has been a considerable consumer of pesticides since 2008. The use of pesticides in Brazil has increased 190% over the last decade, which is significantly above the average global increase of 93% (ANVISA, 2011).

Faria et al. reported that pesticide poisoning was strongly associated with minor psychiatric disorders, emphasizing the dimension of the problem and the importance of adopting new policies for the protection of farm workers' mental health (Faria et al., 1999). Concrete evidence has shown higher numbers of hospitalization due to mood disorders and suicide attempts among residents in areas with a higher use of pesticides (Hou et al., 2016; Kim et al., 2014). On the other hand, these types of studies have controversial results with confounders because hard physical labor, excessive stress and mental suffering are intrinsic to a life of manual labor in the countryside (London et al., 2012b; Wesseling et al., 2010). Considering that intense exposure to pesticides and occupational exposure in agriculture can be risk factors for mental health impairment, studies aiming to investigate the real association between depression and pesticide use are encouraged in different conditions. Defining the direction of the causal link between pesticide use and psychiatric conditions is a significant challenge.

Neurological impairments seem to be associated with the toxicity of different classes of pesticides. Some insecticides can act as cholinesterase inhibitors (Mandour, 2013; Noro et al., 2013) and are well known to be harmful to mental health because of their demonstrated neurotoxicity (Rizzati et al., 2016). In the present study, glyphosate — the most commonly used herbicide worldwide according to the WHO — was by far the most commonly used pesticide in the tillage. Although its use alone was not a significant risk for depressive symptoms, its association with any other pesticide used by rural workers was predictive of an association with a high Beck score. Glyphosate is an

Table 1
Relative, percentage and odds values of univariate analysis between depressive symptoms and demographic characteristics / potential confounding factors (N = 220).

Variables	N	%	OR (95%CI)	p value
Dependent variable: symptoms of depression				
High Beck score (11points and above)	61	27,7%	–	
Low Beck score (0–10 points)	159	72,3%	–	
Variables of Social Demographic characteristics and potential confounding factors				
Age				
18–19	6	2,7%	1.00	–
20–29	35	15,9%	0.59 (0.16, 2.08)	0.415
30–39	54	24,5%	0.51 (0.15, 1.66)	0.267
40–49	59	26,8%	1.10 (0.36, 3.37)	0.860
50–59	48	21,8%	0.90 (0.28, 2.88)	0.871
60 years and above	18	8,2%	1.33 (0.47, 3.73)	0.580
Marital status				
Single never married	30	16,6%	1.00	–
Single separated or divorced	4	2,2%	1.72 (0.21, 14.04)	0.609
Single widowed	0	0,0%	–	–
Married/remarried	139	76,8%	0.67 (0.29, 1.54)	0.351
Cohabiting	8	4,4%	1.03 (0.20, 5.19)	0.965
Brazilian racial categories				
Yellow	7	3,4%	1.00	–
Brown	60	28,8%	1.07 (0.18, 6.04)	0.938
White	124	59,6%	0.94 (0.17, 5.10)	0.947
Black	15	7,2%	1.28 (0.42, 3.93)	0.658
Indigenous	2	1,0%	–	–
Years of schooling				
0 years	10	4,8%	1.00	–
1–4 years	105	50,5%	0.6 (0.15, 2.27)	0.453
5–8 years	41	19,7%	0.62 (0.14, 2.60)	0.514
9–11 years	48	23,1%	0.50 (0.12, 2.07)	0.340
12 years and above	4	1,9%	0.50 (0.03, 6.68)	0.600
Economic status of family				
No salary	46	23,5%	1.00	–
Up to 1 salary	81	41,3%	0.77 (0.34, 1.77)	0.549
1–2 salaries	50	25,5%	1.42 (0.60, 3.38)	0.419
Above 2 salaries	19	9,7%	0.67 (0.18, 2.42)	0.549
Alcohol use	108	53,7%	0.84 (0.46, 1.55)	0.597
Tobacco use	26	13,1%	1.90 (0.81, 4.45)	0.134
Combined alcohol and tobacco use	17	8,5%	1.46 (0.51, 4.15)	0.470
Pesticide exposure	190	86,4%	6.30 (1.45, 27.34)	0.014*
Type of pesticide				
Glyphosate (G)	85	38,6%	0.94 (0.51, 1.73)	0.861
Flutriafol + G	62	28,2%	1.94 (1.24, 3.03)	0.003**
Cyproconazole + G	38	17,3%	2.06 (1.34, 3.17)	0.001**
Thiamethoxam + G	33	15,0%	1.63 (1.10, 2.41)	0.014*
Self-perceived health				
Good	118	58,1%	1.00	–
Poor	85	41,9%	2.48 (1.33, 4.60)	0.004*
Presence of chronic disease	58	27,8%	2.33 (1.22, 4.45)	0.010*

organophosphate compound that has been controversially related to not affect the nervous system at the same intensity as other organophosphorus compounds; however, it has already been described that several herbicides, including glyphosate, may also have important neurotoxic effects (Menendez-Helman et al., 2012; Roy et al., 2016). One of the neurological aggravating factors of pesticide use is suicidal ideation. Suicide *per se* is affected by depression, as well as personality and behavioral traits, such as hostility and impulsivity. Pickett et al. demonstrated that the probability for suicide in a study of Canadian farmers was not higher with insecticide use alone, but it was 71% higher for farmers who used insecticides combined with herbicides

Table 3
Risk factors for depressive symptoms after multivariate analysis (N=220).

High Beck score	OR (95%CI)	p value
Pesticide exposure	5.52 (1.18, 25.88)	0.030
Tobacco use	2.81 (1.11, 7.11)	0.029
Poor self-perceived health	2.61 (1.33, 5.11)	0.005
Presence of chronic disease	2.38 (1.16, 4.87)	0.017

Table 2
Characteristics of the main pesticides used in the farm related by coffee growers.

Name or active ingredient	Class	Chemical name	Empirical formula
Glyphosate	Herbicide	N-(phosphonomethyl)glycine	$C_3H_8NO_5P$
Flutriafol	Fungicide	(RS)-2,4'-difluoro-a-(1H-1,2,4-triazol-1-ylmethyl) benzhydryl alcohol	$C_{16}H_{13}F_2N_3O$
Cyproconazole	Fungicide	(2RS,3RS;2RS,3SR)-2-(4-chlorophenyl)-3-cyclopropyl-1-(1H-1,2,4-triazol-1- yl)butan-2-ol	$C_{15}H_{18}ClN_3O$
Thiamethoxam	Insecticide	3-(2-chloro-1,3-thiazol-5-ylmethyl)-5-methyl-1,3,5-oxadiazinan-4-ylidene(nitro)amine	$C_8H_{10}ClN_5O_3S$

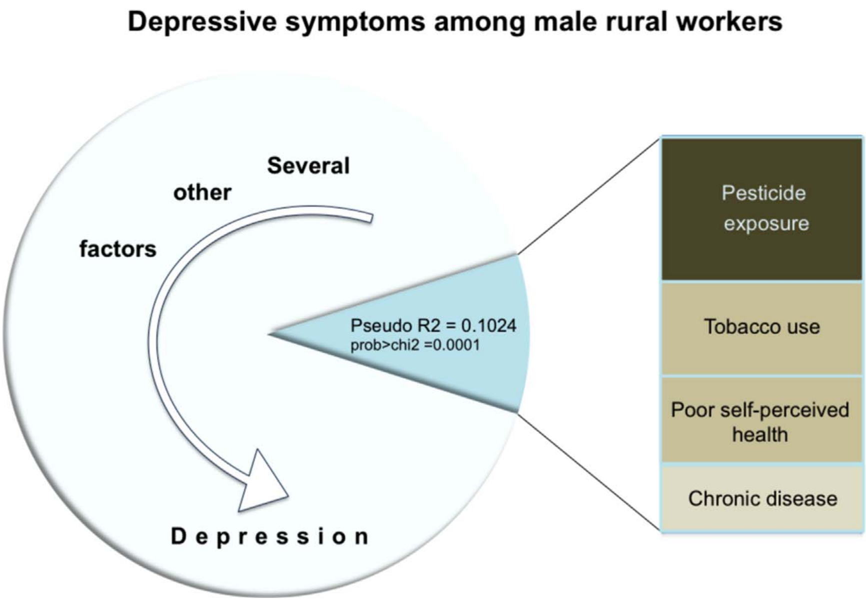

Fig. 3. Results from this study demonstrated some factors associated with depressive symptoms among male rural workers that, summed to other factors, can be focused in actions that aim to improve work and life conditions in the rural zone.

compared with those who were not exposed to these products (Pickett et al., 1998).

Other factors noted in the multivariate analysis of determinant factors for depressive symptoms, according to the BDI, included tobacco use, poor self-perceived health and the presence of chronic disease. The association between smoking and psychiatric comorbidities has been widely studied in different populations (Biggs et al., 2010; Dierker et al., 2015; Dos Santos et al., 2010; Flensborg-Madsen et al., 2011; Khan et al., 2015). Depression is the psychiatric comorbidity that is most commonly associated with nicotine dependence (Pang et al., 2014; Rey et al., 2017). Smoking, as with the use of psychoactive drugs in general, activates the reward brain system, decreasing negative emotional feelings (Durazzo et al., 2013; Weinstein et al., 2010). One limitation of this study is the lack of a diagnosis of tobacco dependence in addition to limited information on the quantity and type of cigarettes smoked because the rural workers were only asked about whether they smoked (yes/no). One future perspective could include an evaluation of the association between the number/type of cigarettes and total Beck score in the multivariate analysis as part of a more precise investigation about smoking and depressive symptoms among rural workers. Individuals with depression tend to have a higher probability of nicotine dependence and a higher level of cigarette smoking (Fergusson et al., 2003).

Variables such as the presence of chronic disease and self-perceived physical health were considered potential confounders in a rural study that highlighted the importance of social determinants of mental health issues, including depression (Liang et al., 2012). Here, these two variables increased the risk of depressive symptoms in the rural population more than two-fold. Interestingly, many studies found that the presence of chronic diseases can lead to bad self-perceived health, resulting in a vicious cycle that leads to depressive symptoms and depression (Lino et al., 2013; Schoevers et al., 2000). On the other hand, depression can also impact self-perceived health status (Deen and Bridges, 2011; Racic et al., 2017), demonstrating that this cycle can go both ways.

Very preliminary biomolecular evidence from our team observed BDNF exon I promoter methylation on this population. The relationship between this epigenetic alteration and depressive disorders has been increasingly described (Castren and Rantamaki, 2010; Kreinin et al., 2015; Su et al., 2016; Yang et al., 2016), suggesting that BDNF can be a peripheral biomarker (Allen et al., 2015). However, the determinant factors and mechanism of this epigenetic change need to be more explored. Factors, such as the quality of the water, the quality of the soil (based on the presence of pesticides or heavy metal), and other life habits and health parameters, such as vitamin D dosage and food and nutritive conditions, have been investigated.

Some limitations need to be considered. The sample includes people from rural communities, even those more remotes. However, it was not possible to recruit people that were not registered in the list provided by the Consortium of the region. Besides, BDI-II was not self-reported when subjects were unable to read (non-schooling, or limited vision). To minimize bias information, the questionnaire was applied in non-public confidential place. Finally, this survey was performed after a period of dry weather and climates conditions can affect mental state of rural workers and consequently the results of studies like this. Future (and on-going) perspectives include keep studying the prevalence of depressive symptoms in the whole region - including men and women - on different moments and also add other variables to better explore risk factors. Overall, the most important goal is working on educational actions that can improve health on such region.

Several factors of mental health were investigated, with a focus on depressive symptoms experienced by rural workers from remote areas in southeast Brazil. This multivariate analysis demonstrates the complexity of finding solutions to improve quality of life in the country, and the use of pesticides remains a relevant factor that needs further guidance. These data contribute to the search for solutions to improve quality of life and mental health in the rural living to the extent that social determinants of depression are being investigated.

Fig. 3 summarizes the factors associated with high Beck scores among male rural workers after multivariate analysis.

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